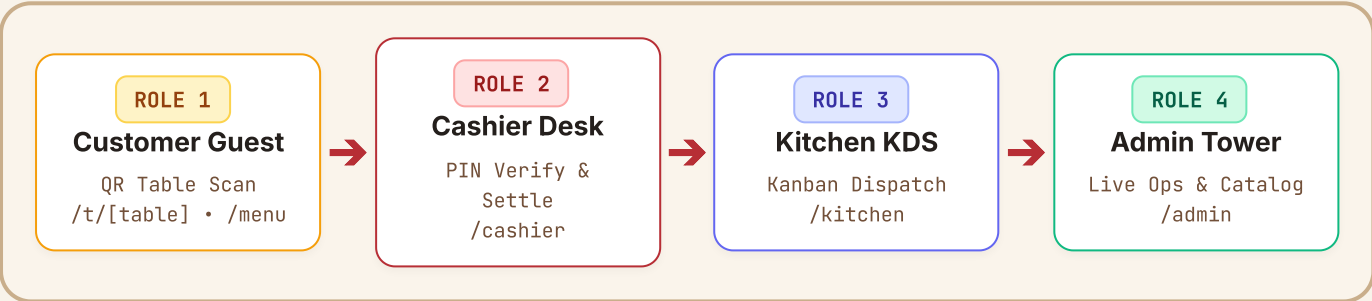


1. Executive Summary & System Purpose

smol café is a state-of-the-art, real-time café operating system and customer dining application. Built with Next.js 15 (App Router), React 19, TypeScript, and Tailwind CSS, it bridges customer self-ordering, front-desk cashier settlement, chef/barista kitchen ticket routing, and owner analytics into a unified real-time synchronization loop.

2. System Architecture & 4 Core Roles



3. The 4 Operational Roles Matrix

Role	Portal URL	Primary Responsibilities	Access / Security
Customer (Guest)	/t/table-XX , /menu	Browse menu, customize drinks, submit orders, track live status, pay via UPI QR.	Public (Bound by Table QR token)
Cashier / POS	/cashier	Verify 4-digit Table PIN, accept orders, manage 12 dining tables, calculate cash change.	Front-Desk Counter Staff
Kitchen (Chef/Barista)	/kitchen	Large-screen KDS Kanban (New → Prep → Ready → Served), acoustic chimes, order timers.	Kitchen Touchscreen Display
Super Admin (Owner)	/admin	Live revenue analytics, hourly demand curves, menu in-stock toggling, loyalty perks.	Secured Gate (PIN: 8888)

⚡ Real-Time Synchronous Engine (lib/sync-events.ts)

Unlike traditional multi-page apps that require frequent polling or refreshes, smol café utilizes an event-driven broadcast channel architecture. Whenever an order is submitted, verified, pushed to

kitchen, or marked ready, events trigger instant UI transitions across all active tabs and devices in under **50ms**.

4. Customer Dining Journey Flow (Patron Dining)

- 1

Table Discovery & Session Binding (/t/table-01 ... table-12)

Guest arrives and scans the cryptographic QR code on their dining table. The URL binds the user's browser session directly to that specific table and seating zone (e.g. *Indoor Cozy, Courtyard Verandah, Garden Terrace, Brew Bar*).
- 2

Menu Browsing & Pure-Code Daily Blackboard (/home , /menu)

Customers browse curated specialty coffees, sourdough melts, fresh bakes, and teas. The homepage features a dynamic, pure-code chalkboard rendered with Caveat handwritten typography showing the daily single-origin roast and recommendation:

today's special: Ratnagiri Estate Pour Over with warm hazelnut notes ♥

— fresh morning bake: cinnamon almond bun ready at 8:00 AM
- 3

Item Customization & Dietary Precision

Clicking any item opens the Customization Drawer: Oat Milk vs Almond Milk vs Whole Milk, sweetness sliders (0%, 50%, 100%), temperature (Hot vs Iced), and barista notes.
- 4

Order Submission & 4-Digit Table Verification PIN (/cart)

When the customer submits their cart, the system assigns a unique Order ID and displays a **4-digit Table Verification PIN**. This prevents prank orders from going to the kitchen until verified by the cashier.
- 5

Live Status Bar & Dynamic UPI Bill Payment (/order/[id] , /bill)

Customers follow the live progress bar: *Order Placed* → *PIN Verified* → *Brewing/Cooking* → *Ready to Serve*. Upon finishing dining, guest requests digital bill, generates a dynamic UPI deep-link QR (`upi://pay?pa=...`), and receives an itemized digital chit with 5% GST calculated.

5. Cashier Desk Front-Line Operations (/cashier)

Tab 1: Verification Queue Incoming orders appear with an amber halo. The cashier asks the patron for their 4-digit PIN, verifies it against the screen, and clicks "Confirm Order". The chit is instantly broadcast to the kitchen.	Tab 2: Tables & Settlement A visual overview of all 12 tables. Cashier can open tables for walk-in guests, view accumulated round totals, launch UPI QR on the counter display, or record Cash payment with automatic change-due calculator.
Tab 3: Paid Orders & Audit Ledger	

Maintains today's complete payment audit trail: Settlement ID, table number, tender method (PhonePe, Google Pay, Cash), exact timestamps, and a button to re-print or view digital guest chits.

6. Kitchen Display System - KDS (/kitchen)

The KDS provides baristas and chefs with an ultra-responsive, touch-optimized Kanban board. Incoming verified chits play an acoustic café chime.

1. NEW ORDERS

Confirmed by cashier.
Waiting for prep.

Action: Start Prep

2. PREPARING

Brewing espresso,
steaming milk, grilling melt.

Action: Mark Ready

3. READY

Plated on counter. Waiter notified.

Action: Serve Table

4. COMPLETED

Successfully served to guest dining table.

Status: Archived

7. Admin Control Tower (/admin)

Protected by an encrypted passcode gate (Demo PIN: 8888), the Admin Tower is the café manager's cockpit:

 Live Sales & Demand Analytics

Real-time daily gross revenue, orders count, table occupancy %, average ticket size (₹384.50), peak demand curve by hour (Peak 5PM: 26 orders/hr), and bestseller rankings.

 12 Tables Floor Map & JSON Tags

Interactive floor plan displaying table status (Free vs Busy). Generates encrypted Table JSON Tags and dynamic QR bill links for any dining table.

 59-Item Menu Stock Toggle

One-click toggling between **IN STOCK** and **SOLD OUT** for all coffees, sandwiches, and treats. Instantly disables sold-out items on active customer devices.

 Smol Club Loyalty & Merchant Setup

Patron point ledger (₹10 spent = 1 pt). Tier perks: Seedling (0-100 pts), Regular (101-500 pts), Smol Insider (500+ pts). Configures merchant UPI VPA and 5% GST rate.

8. Universal Light / Dark Mode & Design Palette

 Daylight Crème vs  Midnight Espresso

Synchronized theme toggle with instant `localStorage: smol_theme` persistence.

Light: #F3E7D3 • #FAF4EB

Dark: #141211 • #1C1917